Experiment No. 16

Footprint a Wireless Network

AIM

To footprint a wireless network by discovering nearby wireless access points and gathering information such as SSID, BSSID (MAC address), channel, signal strength, and encryption type using wireless auditing tools such as Aircrack-ng suite / Wireshark.

PROCEDURE

1. Set the wireless network adapter on the attacker machine into monitor mode using airmon-ng (airmon-ng start wlan0).
2. Use airodump-ng on the monitor-mode interface (airodump-ng wlan0mon) to begin scanning for nearby wireless access points.
3. Observe the list of detected access points, recording each one's BSSID, channel, signal strength (PWR), and encryption/cipher/authentication type (WEP/WPA/WPA2/WPA3).
4. Identify the target access point's SSID and note the list of client devices (stations) currently associated with it.
5. Use a wireless scanning app (e.g., Wi-Fi Analyzer) or Kismet as a cross-check to validate the discovered access point details.
6. Capture a sample of raw 802.11 management frames with Wireshark (using a monitor-mode capture) to inspect beacon frames for additional AP metadata.
7. Compile all discovered access point information, including the target network, into a wireless footprinting report.

SAMPLE OUTPUT

```
$ airmon-ng start wlan0
(monitor mode enabled on wlan0mon)

$ airodump-ng wlan0mon
BSSID              PWR  CH  ENC  CIPHER  AUTH  ESSID
00:14:22:AC:3B:5F  -42  6   WPA2 CCMP    PSK   HomeLab_Target
00:1A:2B:3C:4D:5E  -67  11  WPA2 CCMP    PSK   Office_WiFi
00:1A:2B:3C:4D:6F  -78  1   OPEN -       -     GuestNet

STATION            BSSID              PWR  Probes
A4:5E:60:AB:12:34  00:14:22:AC:3B:5F  -40  HomeLab_Target
```

RESULT

Wireless network footprinting was successfully performed using the Aircrack-ng suite, identifying nearby access points, their SSIDs, BSSIDs, channels, and encryption types, along with associated client stations of the target network 'HomeLab_Target'.